FRAA: A Retrieval-Augmented Agent for Explainable Financial Risk Assessment

Abstract. Financial risk assessment requires probabilities that respond to changing behavior and external evidence while remaining transparent enough for human review. Many deployed systems still depend on engineered tabular features and tree ensembles. These models are effective when feature stores are stable, but they incorporate long behavioral histories and newly released market or regulatory information only indirectly. This paper presents the Financial Risk Assessment Agent (FRAA), a retrieval-augmented model that encodes time-aware user behavior, retrieves timestamp-valid financial documents, fuses the retrieved evidence through cross-attention, and returns both a calibrated risk score and a concise justification. On a proprietary 24-month dataset with over 20 million users, 1.2 billion interactions, and 15 data sources, FRAA achieves a Log Loss of 0.5163 and Recall@5 of 0.4185. This corresponds to a 19.6% Log Loss reduction relative to a deployed Random Forest baseline and a 9.4% reduction relative to a non-retrieval Transformer. Ablation and retrieval analyses indicate that timestamp-valid external evidence is the main source of improvement, with five retrieved documents giving the best balance between useful context and retrieval noise. Expert evaluation rates the generated explanations at 4.32/5 for faithfulness, and the tested serving configuration reaches 15.7 ms latency per query.

Keywords: Financial risk assessment · Retrieval-augmented generation · Explainable AI · Risk scoring · Time-aware Transformer

1 Introduction

Financial institutions assess user-level risk from evidence that is heterogeneous, time-varying, and often incomplete at the moment of decision. Such evidence includes transaction histories, account states, macroeconomic indicators, regulatory disclosures, and news sentiment. Existing risk pipelines often separate behavioral modeling, external evidence collection, and explanation generation into different stages, which makes the final decision difficult to update and audit. Reliable risk estimates support credit review and consumer lending decisions [16,38]. They also underpin fraud prevention and compliance workflows [6,31]. In practice, many deployed systems still use carefully engineered tabular features with tree-based models such as Random Forests [7]. Gradient boosting is another standard choice for tabular risk modeling [13,9]. These baselines remain useful, but the central difficulty is not simply fitting a classifier. It is keeping the risk representation synchronized with behavioral changes and newly released external evidence [11,25].

Neural sequence models offer a more flexible way to model user behavior over time, beginning with recurrent architectures such as LSTM and gated recurrent units [17,10]. Transformer encoders further improve long-range sequence modeling [39]. Sequential recommendation work shows their value for evolving user states [22,36]. Similar temporal modeling ideas have been used in credit-card fraud detection [21]. Yet a purely parametric sequence model remains limited in a financial risk setting. It can use market or regulatory information only if that information was encoded during training or manually attached as a feature. It also gives reviewers little direct evidence for why a high-risk case was flagged. Retrieval methods provide a way to condition decisions on external evidence available at inference time. Classical probabilistic retrieval motivates sparse evidence matching [35]. Dense retrieval and RAG extend this idea to neural document representations and generation [23,26]. Fusion-based retrieval models further show that non-parametric evidence can be integrated into downstream prediction [19]. In parallel, local explanation methods have shown how individual predictions can be inspected [34,29]. Broader XAI surveys emphasize transparency as a system-level requirement [14,4]. Calibration studies show that probability estimates require their own evaluation [15,32]. Post-hoc calibration remains relevant when raw classifier scores are not directly reliable probabilities [40]. These observations motivate a risk model that retrieves time-valid financial evidence and links that evidence to both prediction and explanation.

This work develops the **Financial Risk Assessment Agent (FRAA)**, a retrieval-augmented model for explainable financial risk assessment. The word "agent" is used in a constrained operational sense. For each scoring request, FRAA builds a user state, queries a timestamped financial knowledge base, fuses retrieved evidence with behavioral features, estimates a risk probability, and generates a short explanation grounded in the retrieved context and influential feature groups. FRAA is therefore a structured retrieve-fuse-score-explain pipeline rather than an open-ended autonomous system.

The main contributions are:

- We formulate financial risk scoring as a time-constrained retrieval problem and instantiate it with FRAA, which combines temporal user encoding, timestamp-valid evidence retrieval, cross-attention fusion, and explanation generation.
- We evaluate FRAA on a large-scale proprietary multi-source dataset with a strict chronological split, and compare it with deployed tree-based baselines and neural sequence models.
- We report component ablations, feature-group occlusion, retrieval-depth analysis, explanation quality assessment, and latency measurements to examine the contribution and operational cost of the retrieval and explanation components.

2 Related Work

2.1 Financial Risk Modeling and Tabular Baselines

Classical credit-risk systems are built around statistical classifiers, scorecards, and tree ensembles, which remain strong baselines for heterogeneous tabular data [16,38]. Benchmarking and survey studies show that this pattern holds across credit-scoring datasets, classifier families, and operational fraud settings [3,1]. Later reviews further document the breadth of classification methods used in financial risk modeling [28]. In practice, Random Forests, gradient boosting, and XGBoost remain difficult baselines to displace because they fit mixed tabular features well [7,13]. This is also confirmed by large benchmark comparisons of credit-scoring algorithms [9,25]. Credit and fraud datasets are often imbalanced and loss-asymmetric, so ranking quality and calibration matter alongside raw accuracy [11,8]. Studies of consumer-credit and fraud investigation systems make the same point in realistic review workflows [24,5]. Fraud-detection surveys connect these models to monitoring and compliance settings [6,31]. Their strength, however, also defines the boundary of this paradigm: the model state is largely determined by engineered feature stores, while newly emerging market or regulatory evidence must be manually transformed before it can affect a decision.

2.2 Sequential User Modeling

Sequential recommendation and behavior modeling show that recurrent networks can capture evolving user states [17,10]. Transformer encoders extend this capacity with attention-based long-range dependencies [39]. SASRec and BERT4Rec then demonstrate the value of self-attention for sequential user modeling in recommendation tasks [22,36]. Financial risk assessment has a related temporal structure: transactions, account changes, and market signals form irregular sequences whose relevance depends on timing and context. Sequence models have also been used for credit-card fraud detection, where temporal order is central to identifying abnormal behavior [21]. FRAA follows this line of work but adapts it to calibrated risk scoring, where explanation is part of the target behavior rather than an optional post-processing step.

2.3 Retrieval-Augmented and Explainable AI

Retrieval-augmented methods condition model outputs on non-parametric external evidence. Classical probabilistic retrieval provides the foundation for document matching [35]. Dense passage retrieval and RAG adapt retrieval to neural representations and generation [23,26]. Fusion-in-decoder models show that retrieved evidence can be integrated into downstream generation and prediction [19]. BERT-style encoders and Sentence-BERT provide practical document representations for semantic matching [12,33]. Approximate nearest-neighbor search makes retrieval feasible at large scale [30,20]. In regulated domains, however, retrieval alone is not sufficient. LIME and SHAP support instance-level inspection [34,29]. Broader XAI surveys frame transparency as a modeling and deployment requirement [14,4]. Calibration studies further show that score reliability must be checked separately from discrimination [15,32]. Post-processing methods remain useful when raw scores need probability calibration [40]. Together, these lines of work point to the same requirement: a risk model should expose not only a probability, but also the evidence used to support it.

3 Methodology

3.1 Problem Formulation

For a user u and decision time t, the task is to estimate the probability $\hat{r}_t \in [0, 1]$ that a material adverse event will occur within the next 30 days. The label $y_t \in \{0, 1\}$ indicates whether such an event is observed. The model receives a time-ordered window of behavioral, profile, macroeconomic, and external-document signals available no later than t. This chronological constraint is applied throughout training and testing to avoid future information leakage.

Each training instance is built from an observation window ending at t and a prediction window that immediately follows it. Dynamic events after t are excluded. Slowly changing attributes are read from the latest snapshot available before t. External documents follow the same rule: a document can be retrieved only if its publication or ingestion timestamp is no later than the scoring time. This setup reflects operational risk review, where decisions are made before the adverse event and before later market information becomes available.

3.2 Multi-Source State Construction

FRAA maps the available evidence into three feature families. The dynamic transactional context $\mathbf{d}_t \in \mathbb{R}^{D_d}$ summarizes recent transaction amount, frequency, volatility, merchant category, and event timing. The quasi-static profile context $\mathbf{s}_t \in \mathbb{R}^{D_s}$ contains slowly changing user attributes, including account tenure, product holdings, balance ranges, and risk tier. The market and macro context $\mathbf{m}_t \in \mathbb{R}^{D_m}$ contains time-aligned external indicators available before the scoring time. The external-document branch is motivated by evidence that news and regulatory language can carry risk-relevant signals [37,27].

Let D_d, D_s, and D_m denote the dimensions of the three feature families. These vectors are concatenated and projected into a shared hidden space:

$$\mathbf{x}_t = \mathbf{W}_p[\mathbf{d}_t \| \mathbf{s}_t \| \mathbf{m}_t] + \mathbf{b}_p, \tag{1}$$

where $\|$ denotes concatenation, $\mathbf{W}_p \in \mathbb{R}^{D_h \times (D_d + D_s + D_m)}$ and $\mathbf{b}_p \in \mathbb{R}^{D_h}$ are trainable parameters, and $\mathbf{x}_t \in \mathbb{R}^{D_h}$ is the initial user-state representation.

3.3 Time-Aware Sequence Encoding

Financial behavior is observed at irregular intervals. FRAA therefore injects a time-gap encoding into each event representation. Let $\Delta\tau_i$ denote the elapsed time between consecutive events. The input to layer l of the Transformer encoder is updated as

$$\mathbf{h}_i^{(l)} = \mathrm{TransformerBlock}^{(l)}\left(\mathbf{h}_i^{(l-1)} + \mathbf{p}_{\Delta\tau_i}\right), \quad l = 1, \ldots, L, \tag{2}$$

where $\mathbf{p}_{\Delta\tau_i}$ is the time-gap embedding. The final state $\mathbf{h}_t^{(L)}$ summarizes the user's observed behavior up to time t.

3.4 Retrieval-Augmented Evidence Fusion

To incorporate external knowledge, FRAA converts the encoded user state into a retrieval query:

$$\mathbf{q}_t = f_q(\mathbf{h}_t^{(L)}), \tag{3}$$

where $f_q(\cdot)$ is a lightweight multilayer perceptron that maps the sequence state to the document-embedding space. The financial knowledge base contains approximately 1.2 million timestamped documents, including regulatory filings, market reports, news articles, and macroeconomic commentaries. Documents are embedded offline and indexed with FAISS. At inference time, FRAA retrieves the top-K documents by cosine similarity:

$$\mathcal{D}_t = \mathrm{topK}_{d \in \mathcal{K}_t} \frac{\mathbf{q}_t^\top \mathbf{e}_d}{\|\mathbf{q}_t\| \|\mathbf{e}_d\|}, \tag{4}$$

where $\mathcal{K}_t$ contains only documents available no later than t, and $\mathbf{e}_d$ is the embedding of document d. Document embeddings are produced by a financial-domain text encoder initialized from bidirectional Transformer encoders and sentence-level retrieval models [12,33]. The timestamp restriction prevents future documents from leaking into the prediction.

The retrieved document embeddings are fused with the user state using cross-attention:

$$\tilde{\mathbf{h}}_t = \mathrm{CrossAttn}\left(\mathbf{h}_t^{(L)}\mathbf{W}_Q, \mathbf{D}_t\mathbf{W}_K, \mathbf{D}_t\mathbf{W}_V\right), \tag{5}$$

where $\mathbf{D}_t \in \mathbb{R}^{K \times D_e}$ stacks the embeddings of retrieved documents, D_e is the document-embedding dimension, and $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ are trainable projection matrices. The fused representation $\tilde{\mathbf{h}}_t$ is passed to the scoring and explanation heads. Cross-attention is used instead of simple concatenation because the relevance of external evidence depends on the user state. For example, a macroeconomic commentary may matter for users with concentrated sector exposure, but may be weakly relevant for users whose recent transactions do not involve that sector.

3.5 Risk Scoring and Explanation Generation

The risk head maps the fused representation to a calibrated probability:

$$\hat{r}_t = \sigma(\mathbf{w}_r^\top \tilde{\mathbf{h}}_t + b_r). \tag{6}$$

where $\sigma(\cdot)$ is the sigmoid function, $\mathbf{w}_r$ is a trainable risk-head vector, and b_r is a scalar bias. In parallel, a supervised explanation decoder generates a concise textual justification from the retrieved documents and feature-group importance signals. The training objective combines per-instance binary cross-entropy for risk prediction with an explanation-fidelity term:

$$\mathcal{L} = -y_t \log \hat{r}_t - (1 - y_t)\log(1 - \hat{r}_t) + \lambda \mathcal{L}_{\mathrm{exp}}(\hat{e}_t, e_t^{\mathrm{ref}}), \tag{7}$$

where $\hat{e}_t$ is the generated explanation, e_t^{ref} is an expert or rule-supported reference explanation, $\mathcal{L}_{\mathrm{exp}}$ penalizes mismatches with reference evidence and feature-group rationales, and λ balances prediction and explanation supervision. The reference explanation is constructed from analyst-approved risk factors and timestamp-valid evidence snippets, so the explanation loss encourages the decoder to mention evidence that was actually available at the scoring time. In all reported experiments, the Transformer uses 8 attention blocks, hidden dimension 256, and embedding dimension 512. Retrieval depth is $K = 5$ unless otherwise stated.

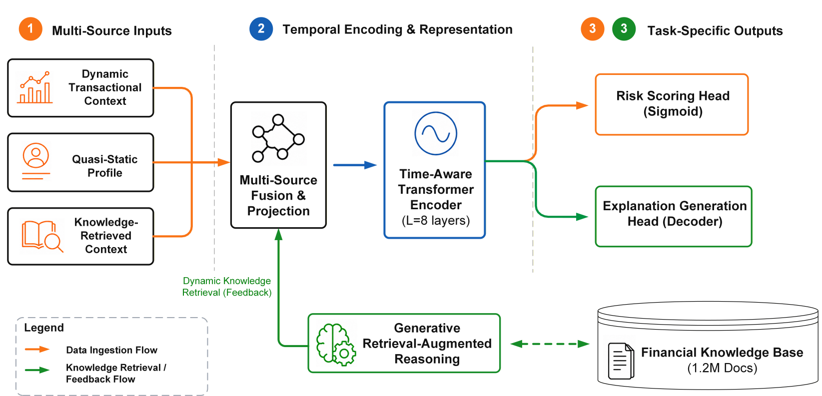

Fig. 1. Overview of FRAA. The model encodes multi-source financial behavior, retrieves timestamp-valid documents from a financial knowledge base, fuses retrieved evidence with the user state, and outputs both a calibrated risk score and an explanation.

Figure 1 summarizes the inference path from multi-source evidence encoding to retrieval, fusion, scoring, and explanation generation.

4 Experiments

4.1 Dataset and Evaluation Protocol

The dataset is a proprietary 24-month multi-source corpus from a financial risk environment. It contains over 20 million users, 1.2 billion interactions, and 15 heterogeneous data sources. Records include transaction events, account status snapshots, product holdings, macroeconomic series, and timestamped news or regulatory signals. The target indicates whether a material adverse event, such as default, severe delinquency, or forced liquidation, occurs within the next 30 days.

We use a strict chronological split. Historical records are used to build lookback features and timestamp-valid knowledge-base entries. The model is trained on a 12-month period, tuned on the following month, and evaluated on the final month. No validation or test-period documents are made available to earlier predictions. We report Log Loss for probability calibration, Recall@1/Recall@5 for risk-review ranking, area under the receiver operating characteristic curve (AUC) for feature-group occlusion, and longest common subsequence ROUGE (ROUGE-L) for explanation overlap with expert-written references.

The evaluation protocol follows the main failure modes of financial risk systems. Log Loss measures whether predicted probabilities are useful for downstream thresholding and risk appetite control. Recall@1 and Recall@5 measure the ranking quality of the highest-risk cases, which are the cases most likely to be reviewed by analysts. The chronological split and timestamp-valid retrieval rule are used for every model variant, including ablations, so that performance differences do not arise from different access to future information.

4.2 Baselines and Training Details

We compare FRAA with a deployed Random Forest (RF), RF with static knowledge-base (KB) features, XGBoost, LSTM, and a non-retrieval Transformer. These baselines cover the main operational choices in the target setting: tabular ensembles, tabular ensembles with manually attached external features, recurrent sequence modeling, and neural sequence modeling without retrieval. We retain the deployed tree-based baseline because credit-scoring studies repeatedly find tree ensembles difficult to displace on heterogeneous tabular data [3,8]. We also include a neural tabular reference point because recent tabular deep learning remains competitive only under careful benchmarking [2]. All baselines use the same chronological split and available feature families. FRAA is implemented in PyTorch and trained with AdamW, learning rate 1×10^{-4}, weight decay 1×10^{-5}, batch size 64, and a maximum budget of 50 epochs. The checkpoint is selected by validation Log Loss, and no test-set information is used for model selection.

Hyperparameters are chosen on the validation month and then fixed for the test month. For retrieval variants, document embeddings are constructed offline and indexed before evaluation. During scoring, the retrieval pool is filtered by timestamp before nearest-neighbor search. This step is essential because a high-capacity retrieval model could otherwise use documents that were unavailable when the decision should have been made.

4.3 Evaluation Questions

The experiments address three questions. First, does retrieval-augmented evidence change calibrated risk prediction and high-risk ranking relative to tree-based and neural baselines? Second, which components and evidence sources account for the observed gains, especially the dynamic transaction encoder, time-gap encoding, and knowledge retrieval module? Third, are the explanations and latency compatible with analyst-facing risk review? The following sections follow this order, separating the main predictive result from auxiliary analyses.

4.4 Main Results

We first compare the full model with deployed and neural baselines under the same chronological split. Table 1 reports the numerical results, and Fig. 2 visualizes the two primary metrics. The ordering is consistent across calibration and ranking: the full retrieval-augmented model outperforms tree-based baselines, the non-retrieval Transformer, and all ablated FRAA variants.

Table 1. Main test-set performance. Lower Log Loss and higher Recall@5 are better.

Model	Log Loss	Recall@5
RF (deployed baseline)	0.6421	0.3172
RF + KB features	0.6289	0.3291
XGBoost	0.6104	0.3495
LSTM	0.5873	0.3738
Transformer	0.5698	0.3846
FRAA (-Dyn)	0.5642	0.3891
FRAA (-TE)	0.5510	0.3957
FRAA (-KB)	0.5385	0.4023
FRAA (full)	**0.5163**	**0.4185**

Relative to the deployed RF baseline, FRAA reduces Log Loss by 19.6% and raises Recall@5 from 0.3172 to 0.4185. Relative to the non-retrieval Transformer, it reduces Log Loss by 9.4% and raises Recall@5 by 8.8%. The ablated variants remain below the full model, indicating that retrieval, temporal encoding, and dynamic features contribute complementary information.

4.5 Ablation and Analysis

Architectural Components The component ablation removes one design choice at a time while keeping the training split, feature availability, and eval-

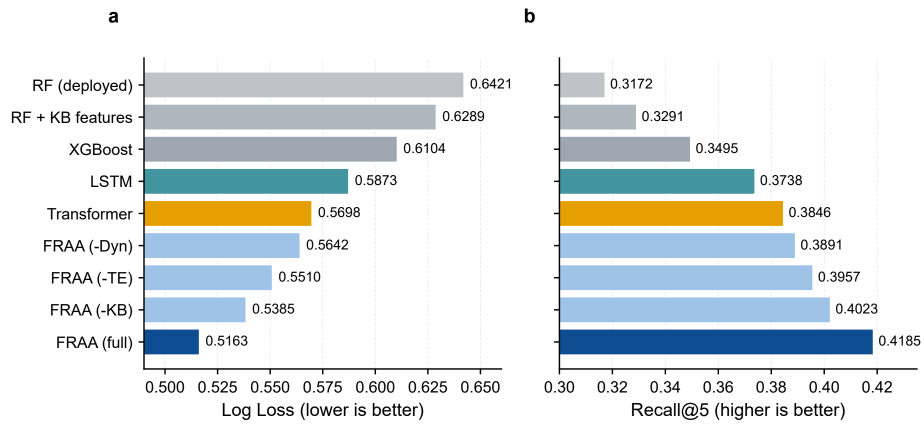

Fig. 2. Main-result comparison. FRAA obtains the lowest Log Loss and highest Recall@5 among all evaluated models.

uation metrics unchanged. This setup separates the effect of retrieval from the effect of stronger sequence modeling.

Table 2. Component ablation on the risk-score prediction task.

Model variant	**Log Loss**	**Recall@5**
Transformer	0.5698	0.3846
FRAA (-Dyn)	0.5642	0.3891
FRAA (-TE)	0.5510	0.3957
FRAA (-KB)	0.5385	0.4023
FRAA (full)	**0.5163**	**0.4185**

Table 2 isolates the contribution of dynamic transactions (-Dyn), temporal encoding (-TE), and knowledge retrieval (-KB). All three removals degrade the full model, confirming that the gains do not come from a single add-on. Removing the knowledge component increases Log Loss from 0.5163 to 0.5385 and decreases Recall@5 from 0.4185 to 0.4023, supporting the retrieval-augmented design. Removing temporal encoding also reduces performance, which is consistent with the irregular timing of financial events. The -Dyn variant remains stronger than the non-retrieval Transformer. External evidence and temporal encoding therefore compensate for some missing dynamic transaction information, but they do not replace it.

Feature Groups and Retrieval Depth We next examine the gain from two complementary views: feature-family occlusion and retrieval depth. Figure 3 summarizes both analyses using the same values as the feature-family occlusion and retrieval-depth experiments. Knowledge-retrieved context produces the largest AUC drop, followed by transaction dynamics. Static profile variables still contribute, but they are less influential than time-sensitive behavioral and external-evidence signals. Retrieval depth also has a clear optimum in this setting: one or three documents provide insufficient external context, whereas seven or ten documents slightly degrade both metrics. We therefore use $K = 5$ in the main experiments. This pattern is expected in retrieval-augmented risk scoring: too few documents leave the model with limited external context, whereas too many documents introduce loosely related evidence that weakens the fused representation.

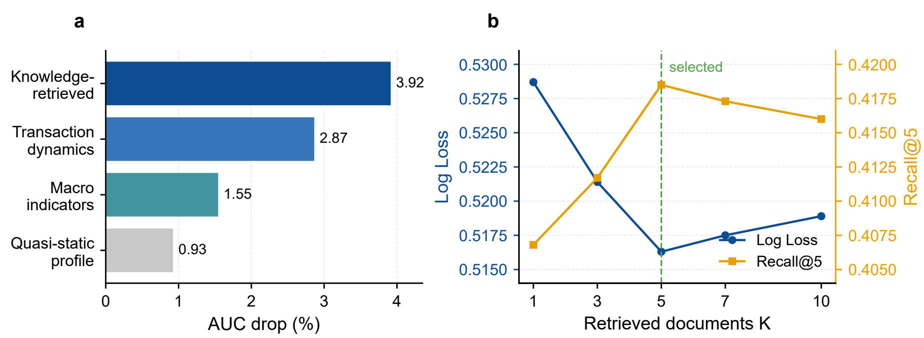

Fig. 3. Retrieval evidence analysis. (a) Feature-family occlusion measured by AUC drop. (b) Retrieval-depth sensitivity, with $K = 5$ selected for the main experiments.

4.6 Further Evaluation

Explanation Quality Beyond risk scores, FRAA is intended to support human review by explaining why a high-risk case was flagged. The goal is not to automate the final decision, but to make the model output easier to inspect. We therefore evaluate the generated explanations using both expert ratings and lexical overlap with reference explanations.

Five domain experts evaluated 1,000 high-risk test predictions. Figure 4 reports higher faithfulness, readability, and ROUGE-L for FRAA than for a template baseline that lists the top three feature groups. The full model obtains faithfulness of 4.32/5, readability of 4.51/5, and ROUGE-L of 0.4131. The result suggests that the explanation head adds information beyond feature enumeration. The expert ratings should be interpreted as an auditability signal, not as evidence that generated text is sufficient for automatic decision making.

Efficiency Analysis The final evaluation asks whether the added retrieval and fusion steps are practical for an analyst-facing review loop.

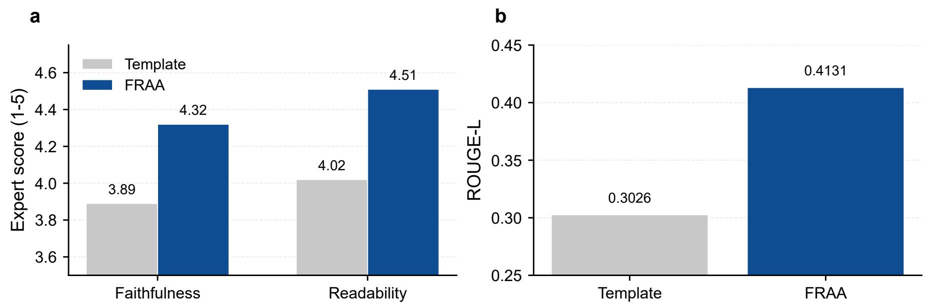

Fig. 4. Explanation-quality evaluation. (a) Expert-rated faithfulness and readability on a five-point scale. (b) ROUGE-L overlap with reference explanations.

FRAA is slower than tree ensembles because it performs retrieval and neural sequence encoding. Even so, the tested latency remains compatible with interactive risk review: FRAA requires 15.7 ms per query and reaches 4,070 queries/s in the batch-size-64 serving configuration. The retrieval step accounts for approximately 4.2 ms of the total latency. Throughput is lower than for tabular baselines, but remains above the level required for batched offline scoring and analyst triage in the tested configuration.

5 Conclusion

This paper presented FRAA, a retrieval-augmented model for explainable financial risk assessment. The main finding is that retrieval helps risk scoring most when external evidence is time-valid and fused with temporal user behavior, rather than attached as a static feature. On a large-scale proprietary multi-source dataset, FRAA obtains the lowest Log Loss and highest Recall@5 among the evaluated models. The ablation, occlusion, and retrieval-depth analyses trace the gains mainly to timestamp-valid external evidence, with five retrieved documents giving the best trade-off between evidence coverage and noise. Explanation and latency evaluations further indicate that the model can support analyst-facing review under the tested serving configuration. The evaluation is limited to one proprietary financial environment, so future work should add public benchmark evaluations where legally possible, repeatability analysis under multiple splits or random seeds, and institution-level transfer tests including parameter-efficient adaptation methods such as LoRA [18]. Taken together, the results suggest that external evidence retrieval is most valuable when it is constrained by time validity, fused with temporal user behavior, and exposed through auditable explanations.

References

1. Abdou, H.A., Pointon, J.: Credit scoring, statistical techniques and evaluation criteria: A review of the literature. Intelligent Systems in Accounting, Finance and Management **18**(2–3), 59–88 (2011). https://doi.org/10.1002/isaf.325
2. Arik, S.O., Pfister, T.: Tabnet: Attentive interpretable tabular learning. Proceedings of the AAAI Conference on Artificial Intelligence **35**(8), 6679–6687 (2021). https://doi.org/10.1609/aaai.v35i8.16826
3. Baesens, B., Van Gestel, T., Viaene, S., Stepanova, M., Suykens, J., Vanthienen, J.: Benchmarking state-of-the-art classification algorithms for credit scoring. Journal of the Operational Research Society **54**(6), 627–635 (2003). https://doi.org/10.1057/palgrave.jors.2601545
4. Barredo Arrieta, A., Diaz-Rodriguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., Garcia, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., Herrera, F.: Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai. Information Fusion **58**, 82–115 (2020). https://doi.org/10.1016/j.inffus.2019.12.012
5. Bellotti, T., Crook, J.: Support vector machines for credit scoring and discovery of significant features. Expert Systems with Applications **36**(2), 3302–3308 (2009). https://doi.org/10.1016/j.eswa.2008.01.005
6. Bolton, R.J., Hand, D.J.: Statistical fraud detection: A review. Statistical Science **17**(3), 235–255 (2002). https://doi.org/10.1214/ss/1042727940
7. Breiman, L.: Random forests. Machine Learning **45**(1), 5–32 (2001). https://doi.org/10.1023/A:1010933404324
8. Brown, I., Mues, C.: An experimental comparison of classification algorithms for imbalanced credit scoring data sets. Expert Systems with Applications **39**(3), 3446–3453 (2012). https://doi.org/10.1016/j.eswa.2011.09.033
9. Chen, T., Guestrin, C.: Xgboost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 785–794 (2016). https://doi.org/10.1145/2939672.2939785
10. Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y.: Learning phrase representations using rnn encoder-decoder for statistical machine translation. In: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing. pp. 1724–1734 (2014). https://doi.org/10.3115/v1/D14-1179
11. Crook, J.N., Edelman, D.B., Thomas, L.C.: Recent developments in consumer credit risk assessment. European Journal of Operational Research **183**(3), 1447–1465 (2007). https://doi.org/10.1016/j.ejor.2006.09.100
12. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 4171–4186 (2019). https://doi.org/10.18653/v1/N19-1423
13. Friedman, J.H.: Greedy function approximation: A gradient boosting machine. The Annals of Statistics **29**(5), 1189–1232 (2001). https://doi.org/10.1214/aos/1013203451
14. Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., Pedreschi, D.: A survey of methods for explaining black box models. ACM Computing Surveys **51**(5), 1–42 (2018). https://doi.org/10.1145/3236009

15. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning. pp. 1321–1330 (2017)
16. Hand, D.J., Henley, W.E.: Statistical classification methods in consumer credit scoring: A review. Journal of the Royal Statistical Society: Series A (Statistics in Society) **160**(3), 523–541 (1997). https://doi.org/10.1111/j.1467-985X.1997.00078.x
17. Hochreiter, S., Schmidhuber, J.: Long short-term memory. Neural Computation **9**(8), 1735–1780 (1997). https://doi.org/10.1162/neco.1997.9.8.1735
18. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: Lora: Low-rank adaptation of large language models. In: Proceedings of the International Conference on Learning Representations (2022)
19. Izacard, G., Grave, E.: Leveraging passage retrieval with generative models for open domain question answering. In: Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics. pp. 874–880 (2021). https://doi.org/10.18653/v1/2021.eacl-main.74
20. Johnson, J., Douze, M., Jegou, H.: Billion-scale similarity search with gpus. IEEE Transactions on Big Data **7**(3), 535–547 (2021). https://doi.org/10.1109/TBDATA.2019.2921572
21. Jurgovsky, J., Granitzer, M., Ziegler, K., Calabretto, S., Portier, P.E., He-Guelton, L., Caelen, O.: Sequence classification for credit-card fraud detection. Expert Systems with Applications **100**, 234–245 (2018). https://doi.org/10.1016/j.eswa.2018.01.037
22. Kang, W.C., McAuley, J.: Self-attentive sequential recommendation. In: Proceedings of the IEEE International Conference on Data Mining. pp. 197–206 (2018). https://doi.org/10.1109/ICDM.2018.00035
23. Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., Yih, W.t.: Dense passage retrieval for open-domain question answering. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing. pp. 6769–6781 (2020). https://doi.org/10.18653/v1/2020.emnlp-main.550
24. Khandani, A.E., Kim, A.J., Lo, A.W.: Consumer credit-risk models via machine-learning algorithms. Journal of Banking and Finance **34**(11), 2767–2787 (2010). https://doi.org/10.1016/j.jbankfin.2010.06.001
25. Lessmann, S., Baesens, B., Seow, H.V., Thomas, L.C.: Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. European Journal of Operational Research **247**(1), 124–136 (2015). https://doi.org/10.1016/j.ejor.2015.05.030
26. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W.t., Rocktaschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive nlp tasks. In: Advances in Neural Information Processing Systems. vol. 33, pp. 9459–9474 (2020)
27. Loughran, T., McDonald, B.: When is a liability not a liability? textual analysis, dictionaries, and 10-ks. The Journal of Finance **66**(1), 35–65 (2011). https://doi.org/10.1111/j.1540-6261.2010.01625.x
28. Louzada, F., Ara, A., Fernandes, G.B.: Classification methods applied to credit scoring: Systematic review and overall comparison. Surveys in Operations Research and Management Science **21**(2), 117–134 (2016). https://doi.org/10.1016/j.sorms.2016.10.001
29. Lundberg, S.M., Lee, S.I.: A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems. pp. 4765–4774 (2017)

30. Malkov, Y.A., Yashunin, D.A.: Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence **42**(4), 824–836 (2020). https://doi.org/10.1109/TPAMI.2018.2889473
31. Ngai, E.W.T., Hu, Y., Wong, Y.H., Chen, Y., Sun, X.: The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. Decision Support Systems **50**(3), 559–569 (2011). https://doi.org/10.1016/j.dss.2010.08.006
32. Niculescu-Mizil, A., Caruana, R.: Predicting good probabilities with supervised learning. In: Proceedings of the 22nd International Conference on Machine Learning. pp. 625–632 (2005). https://doi.org/10.1145/1102351.1102430
33. Reimers, N., Gurevych, I.: Sentence-bert: Sentence embeddings using siamese bert-networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing. pp. 3980–3990 (2019). https://doi.org/10.18653/v1/D19-1410
34. Ribeiro, M.T., Singh, S., Guestrin, C.: "why should i trust you?": Explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 1135–1144 (2016). https://doi.org/10.1145/2939672.2939778
35. Robertson, S., Zaragoza, H.: The probabilistic relevance framework: Bm25 and beyond. Foundations and Trends in Information Retrieval **3**(4), 333–389 (2009). https://doi.org/10.1561/1500000019
36. Sun, F., Liu, J., Wu, J., Pei, C., Lin, X., Ou, W., Jiang, P.: Bert4rec: Sequential recommendation with bidirectional encoder representations from transformer. In: Proceedings of the 28th ACM International Conference on Information and Knowledge Management. pp. 1441–1450 (2019). https://doi.org/10.1145/3357384.3357895
37. Tetlock, P.C.: Giving content to investor sentiment: The role of media in the stock market. The Journal of Finance **62**(3), 1139–1168 (2007). https://doi.org/10.1111/j.1540-6261.2007.01232.x
38. Thomas, L.C.: A survey of credit and behavioural scoring: Forecasting financial risk of lending to consumers. International Journal of Forecasting **16**(2), 149–172 (2000). https://doi.org/10.1016/S0169-2070(00)00034-0
39. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems. pp. 5998–6008 (2017)
40. Zadrozny, B., Elkan, C.: Transforming classifier scores into accurate multiclass probability estimates. In: Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 694–699 (2002). https://doi.org/10.1145/775047.775151